PERSONAL STUDY CARD · NEDALOZANOVSKA · 3:45

Данные и PostgreSQL

Источники → валидация → COPY → схема и SQL

Что нужно знать

- Какой снимок данных используется: sources.json, закреплённая ревизия и SHA-256; чем загруженный файл отличается от Git LFS pointer.
- Единица наблюдения: клуб × сезон × соревнование. Отображаемое имя не является ключом.
- Почему PostgreSQL соответствует фиксированным столбцам и связям; почему COPY быстрее построчных INSERT для этой загрузки.
- Что делает индекс (team_id, season_name, competition_id), почему он создаётся после COPY и зачем нужен ANALYZE.
- Опциональность football-data.co.uk, сопоставление названий клубов, проверка счетов и борьба с дубликатами.
- Лицензия первичного источника пока не подтверждена; публичная скачиваемость не доказывает открытую лицензию.

Что показать

1. Открыть loader/sources.json: ревизия, URL, SHA-256 и порядок столбцов одного источника.
2. Показать ключ агрегации в sql/champagne_queries.sql и объяснить MAX(team_name) как выбор подписи.
3. Показать пример ошибочного счёта/дубликата в tests/test_match_loader.py и ожидаемое поведение.

Короткая речь на английском

> *Our data layer uses eleven pinned CSV tables. We chose PostgreSQL because the records have fixed schemas and are joined by player, club, competition and season. The loader validates headers, uses COPY for bulk import, and builds indices afterwards. We aggregate by canonical identifiers so name aliases do not duplicate observations. An optional match source adds actual scorelines and standings. Invalid or duplicate matches are handled explicitly. The source is publicly downloadable, but its licence is still unconfirmed; we do not claim that this requirement is resolved.*

Переход: “With the data stored consistently, the next step is to define a valid season-to-season target and evaluate it.”

Практические задания

1. Объясни, почему две строки с разными названиями одного клуба нельзя считать двумя наблюдениями.

2. На бумаге проследи 0–0, будущий матч без FTR и два противоречивых счёта одного матча.
3. Запусти тесты загрузчика и объясни причину каждого отказа, не ограничиваясь словом validation.

Вопросы по направлению

Why PostgreSQL rather than MongoDB?

Почему PostgreSQL?

The source tables have fixed columns and relational links by player, club, season and competition. PostgreSQL supports those joins, native COPY and indexed lookups with a clear schema. A different database could work, but this choice follows the data shape.

What identifies one observation?

Что является ключом строки?

For the analytical output it is club ID, season and competition ID. A club's display name is a label, not its identity. We aggregate canonical keys to prevent alias names from creating duplicate observations.

Why build indices after loading?

Почему индексы после COPY?

Maintaining indices during a bulk load adds work to every inserted row. We import with COPY, then build the required indices and ANALYZE the updated tables so the next pipeline has current planner statistics.

Is your source openly licensed?

Подтверждена ли лицензия?

We verified the pinned source revision, headers and checksums, but did not establish an explicit upstream licence. We document this as unresolved and exclude raw data from the repository. Public downloadability alone is insufficient.

What happens if a download is interrupted?

Что будет при обрыве загрузки?

The downloader writes a temporary file and installs it atomically only after checksum and header validation. An interrupted or corrupt download is not accepted as a complete source. Tests exercise these failures using fake HTTP responses.

How do you handle duplicate match rows?

Как обрабатываются дубли матчей?

Identical league/season/date/home/away fixtures are imported once. Conflicting scores for the same fixture stop replacement before stored match rows are changed. Missing optional input clears old enrichment on the next loader run.

What is the main result in one sentence?

Главный результат одним предложением?

The system is reproducible on the measured setup, while historical transfer spending alone has almost no linear relationship with next-season index change and the richer model adds only modest predictive value.

What is the most important limitation?

Главное ограничение?

The data are retrospective and unevenly covered, the index is a proxy, and associations are not causal effects. Source licence and independent team/laptop validation also remain explicit unresolved requirements.

How do goals for, goals against and goal difference work in the comparison?

Как в сравнении рассчитываются забитые, пропущенные голы и разница?

GF and GA are season totals rebuilt from completed home/away match scores. GD is GF − GA. They come from the optional football-data.co.uk match enrichment, not from the Champagne Index player totals.

How are attacking and defensive ranks calculated?

Как рассчитываются атакующий и защитный рейтинги?

Within one league and season, attack rank orders goals scored per match from highest to lowest; defense rank orders goals conceded per match from lowest to highest. SQL RANK gives tied rates the same rank and leaves a gap after ties. They describe scores, not tactics or chance quality.

Do we have xG or can we estimate it?

Есть ли в данных xG или можем ли мы его оценить?

The source has no provider xG field. The comparison instead shows observed shots and shots on target. A separate estimate multiplies shots on target by the league-season average goals-per-shot-on-target rate; it is an experimental volume proxy with no shot-location or chance-quality input, not xG.

What is the difference between the Champagne Index and attack rank?

Чем атакующий рейтинг отличается от Champagne Index?

The Champagne Index is recorded player goals plus assists divided by a Transfermarkt matches proxy for a club-season. Attack rank uses actual completed league scorelines, sums team goals and ranks goals per match within the league-season. Their sources, coverage and units differ.

How did you check the goalkeeper defensive totals?

Как вы проверили вратарские показатели по пропущенным?

We compared Transfermarkt goalkeeper goals-conceded totals with the independent match-result totals on 388 adequately covered top-five club-seasons. They match exactly in 366 (94.3%). Clean sheets match only 269 (69.3%), so the scoreline source remains the basis of league ranks.

What does the defensive-spend analysis show?

Что показывает анализ расходов на защитников?

The Pearson correlation between defender transfer fees and next-season change in the goalkeeper-source conceded-goal rate is r=0.004047, p=0.426997, n=38,531. The sample does not support a useful linear association. It is descriptive, not causal. A separate match-level check finds teams with a red card concede 0.341 more goals on average in the same match, but match context confounds that association.

Why keep the player-source defensive values separate from scoreboard goals?

Почему defensive-показатели по игрокам отделены от итогов матчей?

The sources use different records and denominators. Goalkeeper conceded totals agree with scorelines in 94.3% of adequately covered seasons, while clean sheets agree in 69.3%. The UI therefore uses match scores for GF, GA, GD and ranks; player-source totals remain a quality cross-check. Shots and corners are also observed match statistics, not chance-quality measures.

Is the shot-based estimate the same as xG?

Оценка по ударам — это xG?

No. It multiplies a team's shots on target by the league-season average goals per shot on target. It has no shot-location, distance, body-part, pressure or goalkeeper-position input. It is a rough volume benchmark and must be called an SOT-based proxy, not provider xG.

What does the Poisson attack/defense rank tell us?

Что показывают Poisson-ранги атаки и обороны?

The model fits attack, defense and a shared home-advantage term to the selected league-season scores. Larger fitted attack and defense parameters mean stronger estimated attack and defense. Its ranks are descriptive and in-sample; the separate chronological RandomForest evaluation is the project's held-out prediction check.

Did the added defensive features improve the model?

Улучшили ли защитные признаки модель?

On the same chronological split, R^2 moves from 0.1271 with the five-feature baseline to 0.1502 after adding goalkeeper conceded rate, discipline and net spend; MAE moves from 0.4856 to 0.4788 on 5,585 held-out rows. The gain is modest, and it does not isolate a causal effect of defense.

Do red cards cause more goals to be conceded?

Красные карточки приводят к большему числу пропущенных?

The local sample shows a same-match difference of +0.341 conceded goals per team-match (1,619 sent-off vs 16,195 others; Welch $p < 0.0001$). This is observational: stronger opponents, match state and timing can affect both red cards and goals. It is not a causal estimate.

Какие фразы не использовать

- Все источники имеют открытую лицензию.
- Индекс ускоряет любой запрос в 145 раз.
- Ноль в данных всегда означает подтверждённое отсутствие события.

> Роль — учебная ответственность. Не заявляй личное авторство или самостоятельную проверку, если этого не было. Покажи ровно те изменения и команды, которые выполнял(а) лично.